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Hiring Team
Silicon Quantum Computing
Sydney, NSW

Dear Hiring Team,

I am writing to apply for the Quantum Engineer - Control and Automation role at Silicon Quantum Computing. I am especially interested in the role because it combines quantum hardware, measurement automation and software engineering: an environment where careful experimental workflows, reliable Python tooling and hardware-aware analysis directly increase processor uptime and measurement throughput.

My background is in electronic engineering, embedded software and RF/DSP research. I have 4+ years of experience developing C/C++ and Python software close to hardware, and I am completing a master's degree focused on RF power-amplifier behavioural modelling, Digital Predistortion and FPGA-oriented implementation. In that research, I develop Python reference models, analyse measurement-driven datasets, evaluate performance metrics such as NMSE, and build repeatable validation flows that compare algorithmic models, fixed-point representations and VHDL simulation outputs. This experience maps well to calibration and tune-up work where measurement quality, repeatability and clear metrics are central.

At IoTag Technology, I worked as an Embedded & Software Engineer on telemetry and equipment-monitoring systems. I built data acquisition and communication pipelines between devices, sensors and backend systems, created Python tooling for validation and reliability checks, and debugged hardware/software issues across deployed embedded systems. Earlier, at Lactec, I worked on embedded measurement systems and capacitive sensor prototypes, supporting data acquisition, validation and hardware/software integration. These roles strengthened my ability to turn device-level behaviour into maintainable software workflows and to work across hardware, software and analysis boundaries.

I have not yet worked directly with spin qubits, but I bring a strong foundation in electrical engineering, RF measurement concepts, Python-based analysis, automation and experimental validation. I am highly motivated to apply that foundation to quantum device characterisation, automated calibration and control-stack integration, and to learn from SQC's physicists, quantum engineers and software teams. The opportunity to help make quantum processors run themselves is exactly the kind of multidisciplinary engineering challenge I am looking for.

I am based in Sydney and would welcome the opportunity to discuss how my Python, embedded systems, RF/DSP and measurement-automation background could contribute to SQC's Control and Automation team.

Sincerely,

Leonardo De Andrade Santos